

PROJECT REPORT

Harnessing EC2, VPC & MobaXterm for Cutting-Edge Entity Hosting

1. Introduction

Cloud computing has revolutionized the way organizations deploy and manage their IT infrastructure. Instead of relying on traditional physical servers, businesses are increasingly shifting towards cloud-based solutions that offer scalability, flexibility, and cost efficiency. Amazon Web Services (AWS) is one of the leading cloud service providers that offers powerful tools such as Elastic Compute Cloud (EC2) and Virtual Private Cloud (VPC) to build secure and scalable systems. In this project, we focus on harnessing EC2 for virtual server deployment, VPC for secure network configuration, and MobaXterm as a remote access tool. This combination enables users to efficiently host applications, manage servers remotely, and maintain a secure environment without the need for heavy infrastructure investment.

2. Objectives

The primary objective of this project is to design and implement a cloud-based hosting solution using AWS services. It aims to deploy applications using EC2 instances while ensuring secure communication through VPC configuration. Another objective is to enable remote server management using MobaXterm via SSH protocol. The project also focuses on improving scalability and reliability while reducing operational costs. Additionally, it aims to provide a user-friendly and efficient hosting environment suitable for developers, startups, and organizations.

3. Problem Statements

In the modern digital environment, many users face challenges in deploying and managing applications due to a lack of proper infrastructure. Traditional hosting methods require significant investment in hardware and maintenance, making them less feasible for small businesses and individuals. Moreover, secure remote access to servers is often difficult without proper tools and configurations. Another major issue is the absence of isolated network environments, which increases the risk of security breaches. Scalability is also a concern, as traditional systems cannot handle sudden increases in traffic efficiently. These problems highlight the need for a cloud-based solution that is secure, scalable, and easy to manage.

4. System Requirements

The system requires basic hardware such as a computer or laptop with a stable internet connection. On the software side, an AWS account is necessary to access cloud services like EC2 and VPC. A web browser is required to access the AWS Management Console, and

MobaXterm is used as a remote access tool for connecting to the EC2 instance. The system can run on operating systems such as Windows or Linux, making it accessible to a wide range of users.

5. System Architecture

The system architecture consists of several key components that work together to provide a complete hosting solution. The EC2 instance acts as a virtual server where applications are deployed. The VPC provides a secure and isolated network environment, allowing users to control network configurations such as IP ranges and subnets. Security groups act as virtual firewalls to control inbound and outbound traffic. An internet gateway is attached to the VPC to enable communication with external networks. MobaXterm is used to establish a secure SSH connection to the EC2 instance, allowing users to manage and configure the server remotely.

6. Modules of the System

The system is divided into multiple modules to ensure efficient functionality. The EC2 Instance Module is responsible for launching and managing virtual servers. The VPC Module handles the creation and configuration of the virtual network. The Security Module ensures that only authorized users can access the system by configuring security groups and firewall rules. The Remote Access Module uses MobaXterm to establish secure SSH connections to the server. Finally, the Deployment Module is responsible for installing necessary software and deploying the application on the server.

7. Implementation

The implementation of the system begins with creating an AWS account and accessing the EC2 dashboard. A new EC2 instance is launched by selecting an appropriate Amazon Machine Image (AMI) and instance type. Next, a VPC is configured with subnets, route tables, and an internet gateway to enable connectivity. Security groups are set up to allow SSH access through port 22. The user then connects to the EC2 instance using MobaXterm by providing the public IP address and the key pair file. Once connected, the necessary software such as a web server is installed, and the application is deployed. Finally, the application is tested by accessing it through a web browser.

8. Advantages

The system offers several advantages over traditional hosting methods. It provides scalability, allowing users to increase or decrease resources based on demand. It is cost-effective, as users only pay for the resources they use. The system ensures security through VPC and security groups. Remote accessibility is another key benefit, as users can manage the system from anywhere using MobaXterm. Additionally, the system offers high availability and flexibility, making it suitable for various applications.

9. Limitations

Despite its advantages, the system has some limitations. It requires a stable internet connection to function properly. Users may face a learning curve when working with AWS services for the first time. The initial setup process can be complex, especially for beginners. There is also a risk of security misconfiguration if proper precautions are not taken while setting up the system.

10. Future Enhancements

The system can be further improved by implementing advanced features such as auto-scaling, which automatically adjusts resources based on demand. Load balancing can be added to distribute traffic evenly across multiple servers. Integration with database services like Amazon RDS can enhance data management capabilities. Monitoring tools such as CloudWatch can be used to track system performance and detect issues in real-time. Additionally, implementing CI/CD pipelines can automate the deployment process and improve efficiency.

11. Conclusion

In conclusion, this project successfully demonstrates the use of AWS services for building a secure and scalable hosting environment. By leveraging EC2, VPC, and MobaXterm, the system provides an efficient solution for deploying and managing applications. It eliminates the need for physical infrastructure while ensuring high performance and security. The project is highly beneficial for developers, startups, and organizations looking to adopt cloud computing for their operations.